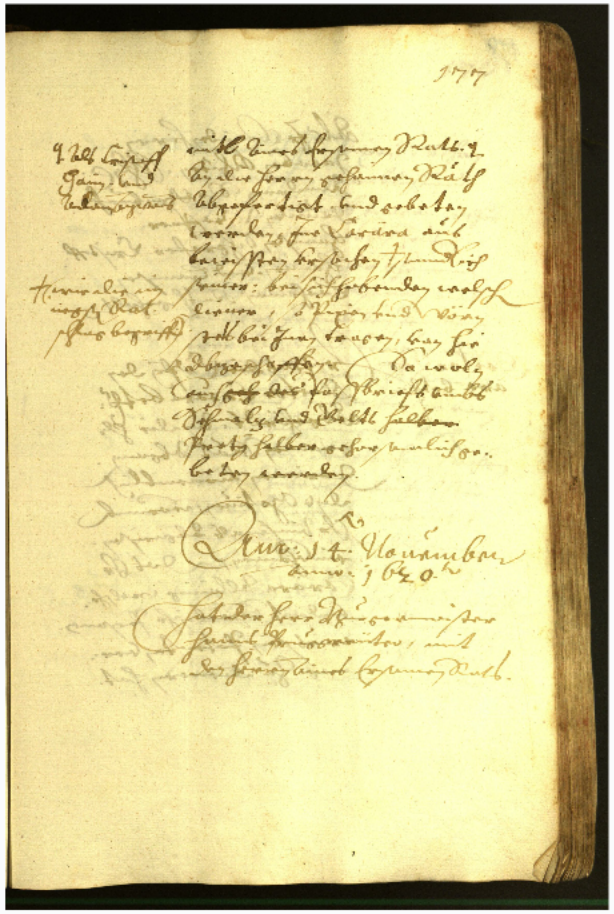
FYP RESEARCH PROGRESS · 27 SEPTEMBER 2026

Active learning for historical OCR

Can better page selection reduce the labelled data needed to adapt small vision-language models?

Evidence through EXP-013 · READ2016 / Bozen

Dataset and the unit of annotation



Historical German handwriting

Official split	Pages	Text lines
TRAIN pool	350	8,367
VAL	50	1,043

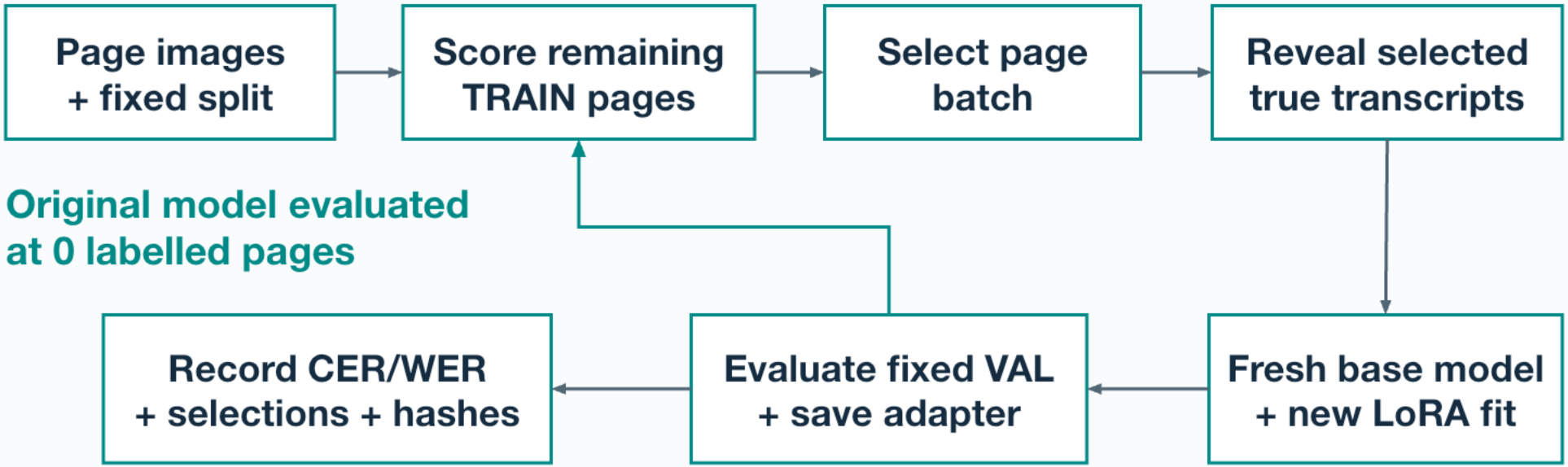
Select one TRAIN page → reveal every line transcript

Existing true labels simulate annotation.
Model scoring sees images; VAL truth stays with the evaluator.

8 VAL pages for recipe checks · 42 fixed VAL pages for reported curves

Source: READ2016 / Bozen v1.2.0 · Toselli et al. · CC BY 4.0

The active-learning pipeline



Each round fits all acquired labels; later uncertainty uses that arm's preceding adapter.

TRAIN oracle ≠ VAL evaluator · Random skips scoring · Stop after 3 fixed rounds

Why the first joint OCR recipe was paused



What we established

Training, checkpoint reload and evaluation eventually completed. We also tried stronger fitting and simpler box/text output.

What prevented a useful comparison

Outputs stayed invalid, repetitive or capped. CER stayed near 1; line-box F1 stayed at 0. Simplifying JSON alone did not fix recognition.

Research decision: isolate transcription before testing selection

Poor usable OCR masks acquisition effects. This rejects the tested recipe, not all joint OCR.

Evidence: grouped reader reports in results/EXP-001-002.md, EXP-003-005.md, EXP-006-008.md

Can historical page transcription learn?

Question: does plain page text learn before we compare selectors?

EXP-009 · LightOnOCR-2-1B-base

Compact OCR specialist

Built for page-image transcription; the author recommends this base variant for domain fine-tuning.

EXP-010 · Qwen3-VL-4B-Instruct

Small general-purpose VLM

Advertises document OCR capability. Keeping Qwen tests whether simplifying the output task can unlock learning.

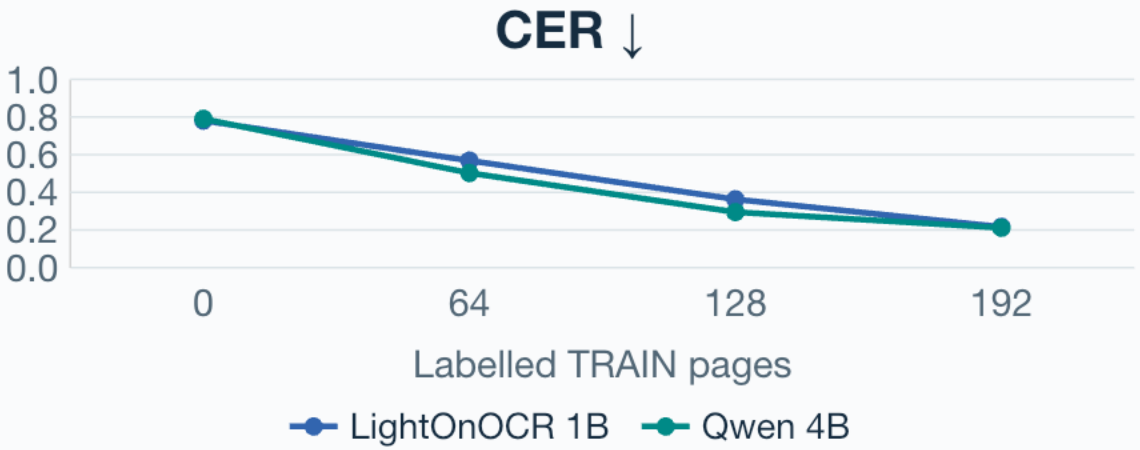
Shared numeric recipe: random 0 → 64 → 128 → 192 pages; seed 824

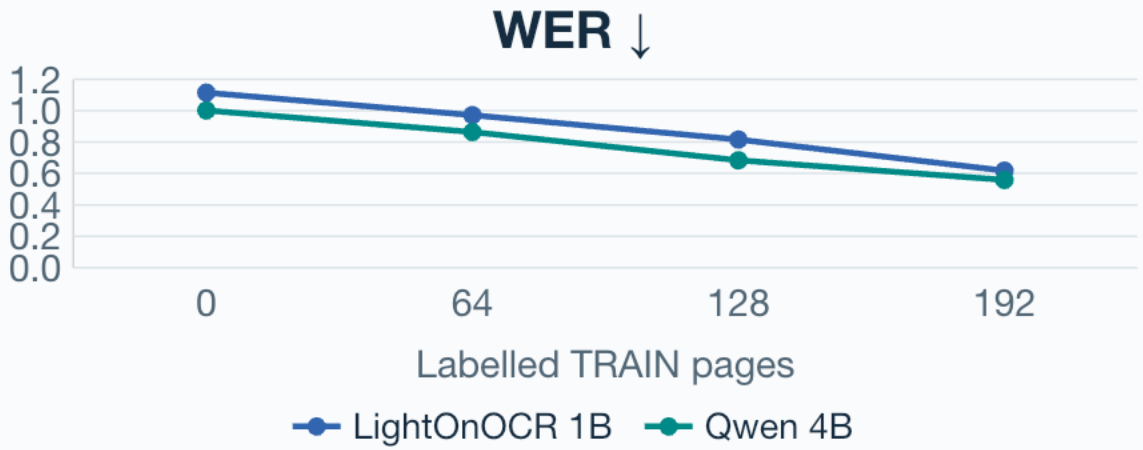
3 epochs per fresh cumulative fit · LoRA r8 / α16 · LR 10^{-4} · HF Transformers + PEFT

Specialist versus general VLM is an engineering comparison; processors and prompts differ.

Model sources: LightOnOCR base card / report · Qwen3-VL-4B-Instruct card / report

Both models learn under random selection





Pages	LightOn CER	LightOn WER	Qwen CER	Qwen WER
0	0.7812	1.1155	0.7902	1.0019
64	0.5683	0.9720	0.5025	0.8643
128	0.3625	0.8164	0.2946	0.6841
192	0.2161	0.6179	0.2112	0.5589

~72–73% relative CER reduction at 192 pages; substantial word errors remain.

Same fixed VAL42 · CER/WER are edit counts divided by reference length; lower is better

Entropy and confidence estimate OCR uncertainty

$p_t(v)$: raw next-token probability · y_t: generated token · T: tokens · V: vocabulary size

Entropy

$$H(\text{line}) = -\Sigma_t \Sigma_v\, p_t(v) \log p_t(v) \,/\, (T \log V)$$

Select the highest page H: token distributions have more competing possibilities.

Least confidence

$$C(\text{line}) = \exp\left[\, \Sigma_t \log p_t(y_t) \,/\, T \,\right]$$

Select the lowest page C: the generated transcription has lower token confidence.

Random control

Seeded uniform page selection without replacement; no model score.

EXP-011: one page sequence. EXP-013: average 3 fixed line scores into each page score.

Uncertainty motivation: Settles (2009), §3.1 · formulas below are our generated-token adaptation

K-center expands frozen visual coverage

z: frozen normalized page feature · L: acquired pages · U: remaining pages

$$d(x, L) = \min_{l \in L} \|z(x) - z(l)\|_2$$

Choose $x^* = \operatorname{argmax}_{x \in U} d(x, L)$; add x^* to L; update distances; repeat.



EXP-011: begin from the shared 64 selected pages.

EXP-013: with L empty, first choose the page nearest the pool's mean feature.

Why: cover different inputs without relying on text confidence; visual distance may miss OCR difficulty.

Coverage motivation: Sener & Savarese, ICLR 2018 · greedy Euclidean coverage in frozen features

Whole-page selection after a shared warm start



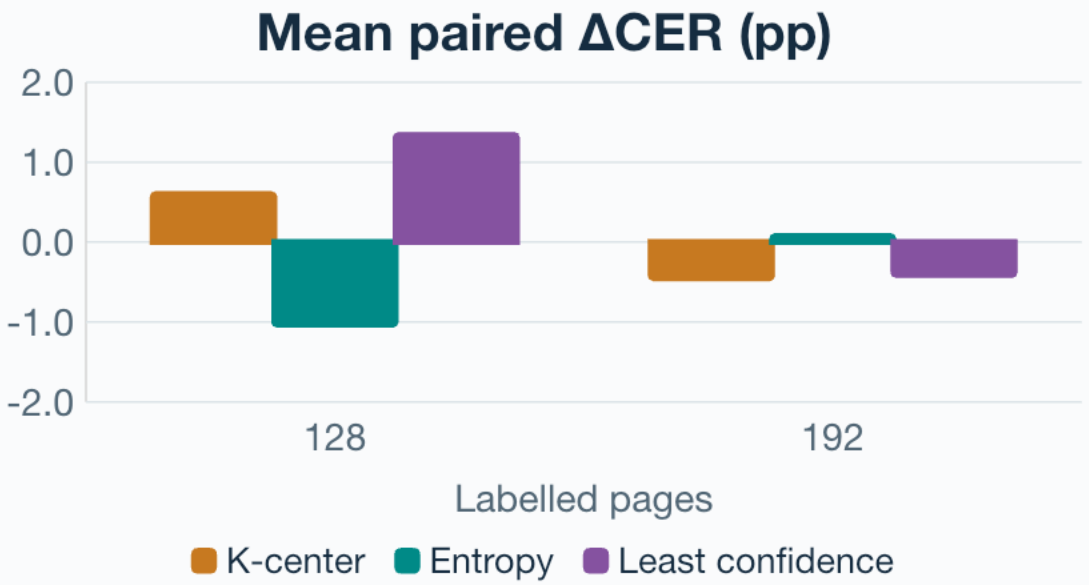
Purpose: test which remaining pages help, after the same initial supervision.

Strategy	128 CER	128 WER	192 CER	192 WER
Random	0.2856	0.6720	0.2096	0.5550
K-center	0.2917	0.6865	0.2051	0.5554
Entropy	0.2753	0.6709	0.2104	0.5533
Least conf.	0.2991	0.6834	0.2055	0.5543

All arms: 0-page CER/WER 0.7902 / 1.0019; mean at 64: 0.4996 / 0.8637.

Entropy leads mean CER at 128, but the later ranking changes; paired gaps tell the fuller story.

Whole-page selection gains vary across seeds



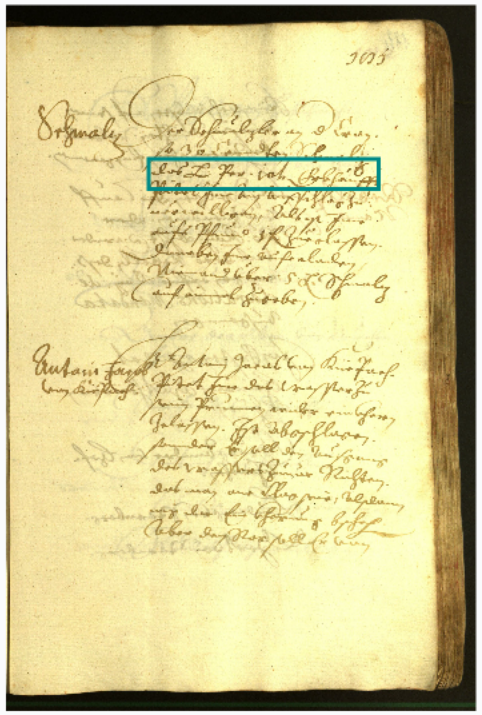
Method / pages	ΔCER ± SD	ΔWER ± SD
K-center 128	+0.60 ± 1.07	+1.45 ± 0.21
K-center 192	−0.45 ± 0.85	+0.04 ± 1.00
Entropy 128	−1.03 ± 0.69	−0.11 ± 0.82
Entropy 192	+0.08 ± 2.18	−0.17 ± 1.44
Least conf. 128	+1.34 ± 3.85	+1.13 ± 1.58
Least conf. 192	−0.41 ± 2.48	−0.07 ± 1.31

No selector consistently wins both budgets across CER and WER.

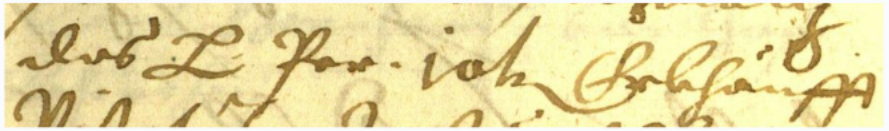
Seed 824: 26 pool outputs hit the cap yet had low entropy / high confidence.
Uncertainty can miss repetitive failures; uncapped-page checks changed some rankings.

Δ = selector − same-seed random · percentage points · negative values favour the selector

Known boxes isolate text recognition



Whole page



Supplied line box → Qwen transcribes only this crop

Short text outputs in source reading order

Why this helps

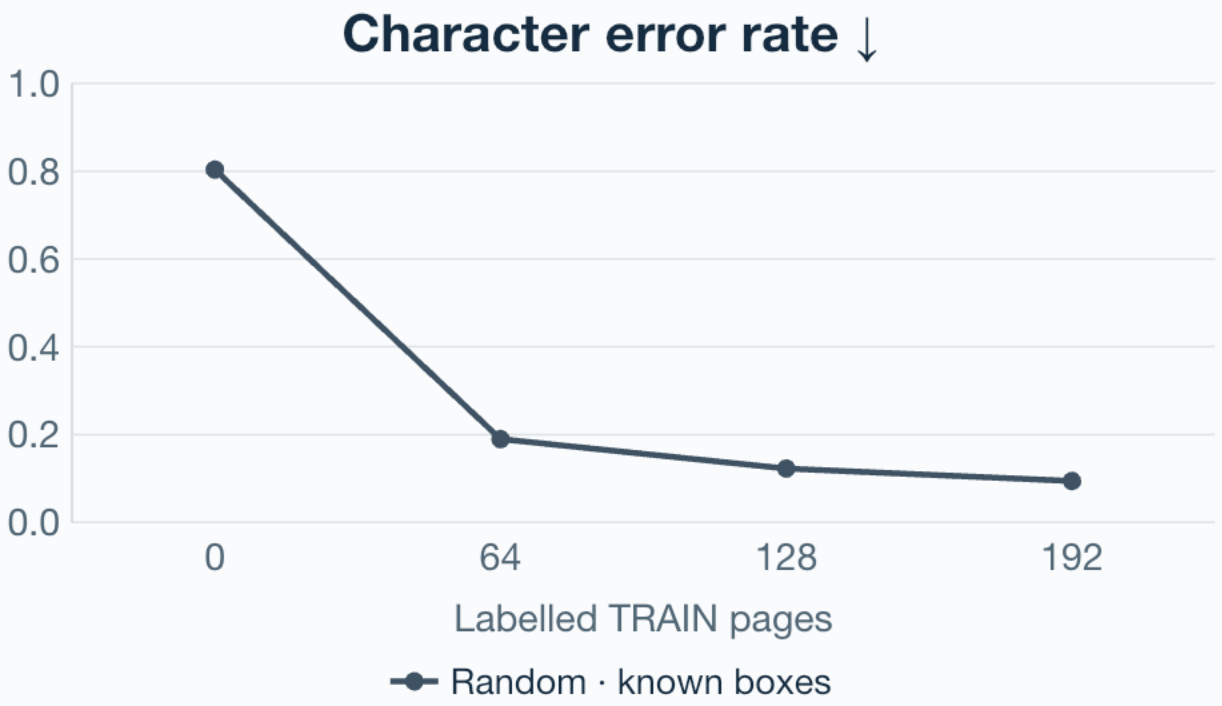
Separate recognition from localization.

Reduce long-output failures.

Study selection with usable OCR.

This is a different task and recipe; its CER is not a fair head-to-head gain over whole-page OCR.

Random selection establishes a known-box baseline



Pages	Train lines	CER	WER
0	0	0.804	1.087
64	1,537	0.190	0.581
128	3,037	0.123	0.452
192	4,567	0.094	0.370

The largest drop comes with the first 64 pages

1 epoch per fresh fit · 3 fixed rounds · capped VAL regions: 2 → 0 → 0 → 0

Establishes adaptation and a control curve; no model-informed selection comparison yet.

Source: results/EXP-012.md · same 42 VAL pages / 861 lines at every checkpoint

Strategy-first selection at smaller page budgets

Why smaller budgets?

EXP-012's largest gain came early.
24-page batches leave more headroom
to test the earliest selection effects.

Why remove the warm start?

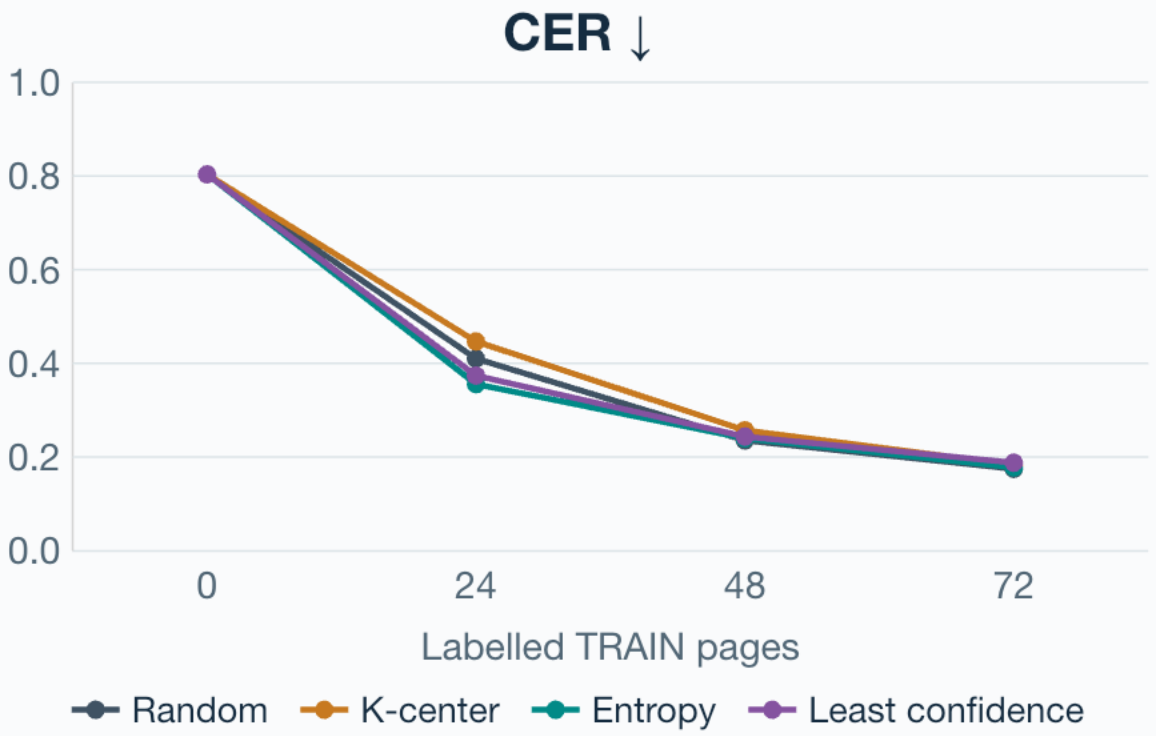
EXP-011 tested additions after 64 labels.
Now each strategy chooses its own
first 24 from the original model.

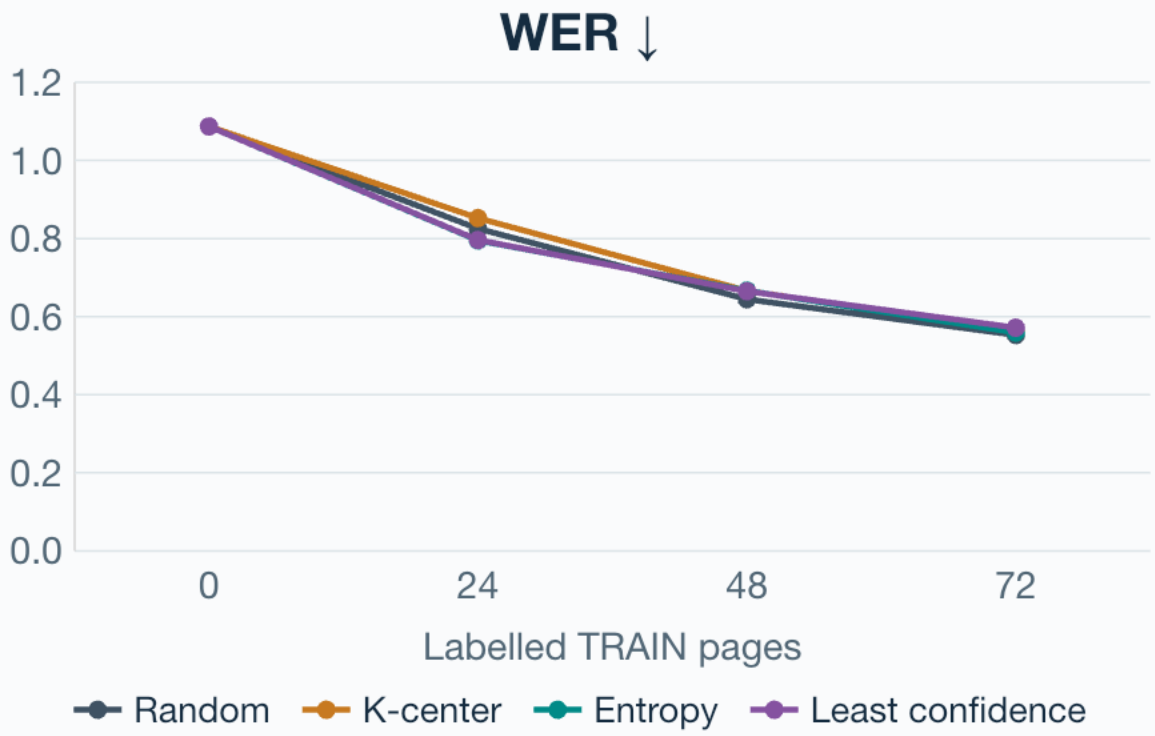


Score lines at ¼, ½, ¾ of reading order; selection reveals all page labels.
Later uncertainty uses its own adapter; k-center features stay frozen.

0-page VAL42 baseline reused from EXP-012 after identity checks and metric recomputation.

Uncertainty leads early; random leads later





Early lead: entropy reduces CER versus random by 5.45 percentage points at 24.

The lead is not sustained; this is one pilot seed, not proven annotation savings.

Same VAL42 / 861 supplied lines · original-model zero baseline reused from EXP-012

RESULTS TABLE · RATES, NOT PERCENTAGES · LOWER IS BETTER

EXP-013: CER and WER at every page budget

Character error rate (CER)

Strategy	0*	24	48	72
Random	0.8036	0.4104	0.2357	0.1750
K-center	0.8036	0.4469	0.2574	0.1838
Entropy	0.8036	0.3559	0.2400	0.1781
Least conf.	0.8036	0.3740	0.2439	0.1881

Word error rate (WER)

Strategy	0*	24	48	72
Random	1.0871	0.8254	0.6445	0.5529
K-center	1.0871	0.8523	0.6662	0.5664
Entropy	1.0871	0.7951	0.6665	0.5593
Least conf.	1.0871	0.7963	0.6650	0.5716

24 pages: entropy leads both metrics; confidence also improves over random.

72 pages: entropy is close, but random has the lowest CER and WER.

*0 is reused, not a new run. Capped VAL lines: 2/861 at 0; 0/861 at every trained checkpoint.

Values rounded to 4 decimals; exact metrics and receipts are preserved in results/EXP-013.md

Uncertainty strategies select overlapping pages

Cumulative page intersection

Strategy pair	24	48	72
Random / K-center	2	6	13
Random / Entropy	2	6	14
Random / Confidence	2	5	17
K-center / Entropy	1	5	16
K-center / Confidence	1	9	20
Entropy / Confidence	16	24	37

Cumulative training lines

Strategy	24	48	72
Random	589	1,155	1,725
K-center	501	1,071	1,658
Entropy	580	1,160	1,713
Least conf.	584	1,116	1,659

Entropy/confidence share 37 of 72 pages. Different scores can yield similar batches.

Equal page budgets do not mean equal transcript volume or equal human annotation effort.

Pair overlap = shared acquired TRAIN pages · line counts = all labels revealed by those pages

Current findings and the next research test

Established

Domain fine-tuning improves recognition.

Both page and crop pipelines run.

Random is a necessary strong control.

Still unproven

Repeatable annotation savings.

Independent-document generalization.

Box detection or human-time savings.

EXP-013: one seed; LoRA initialization was not explicitly seeded. Official validation only.

Next: repeat the 24/48/72 comparison with controlled LoRA initialization

Seed before constructing adapters; pair multiple seeds; retain random and fixed evaluation.
Then test whether OCR uncertainty plus diversity improves early-budget selection.

Methodological context: Munjal et al., CVPR 2022 · historical recognition: Aradillas et al., 2020

Sources that motivate the research design

Settles · Active Learning Literature Survey (2009)

Classical uncertainty sampling; our token/page scoring is an adaptation.

Sener & Savarese · ICLR2018

Core-set coverage motivates the greedy k-center baseline.

Munjal et al. · CVPR2022

Strong random controls and stochasticity-sensitive evaluation.

Aradillas, Murillo-Fuentes & Olmos · 2020 preprint

Historical handwriting recognition with few labelled lines.

Bai et al. · Qwen3-VL (2025) / Taghadouini et al. · LightOnOCR (2026)

General VLM and compact OCR-specialist model context.

Toselli et al. · READ2016 / Bozen v1.2.0

Page images and line annotations; CC BY4.0.

Experiment results index: grouped EXP-001–008 and individual EXP-009–013 reports